NMDC DREDGING & MARINE	NMDC GROUP P.J.S.C SURVEY DEPARTMENT			
		Project No.		YYYY.WW
	Weekly Survey Report	**20012**	**WSR**	2026.43

Project name:	**Tung-Hsiao Power Plant 2nd Stage Renewal Project Turn-key Contract of Subsea Gas Pipeline**
Name:	Daryll Enano
E-mail:	daryll.enano@nmdc-group.com
Date:	October 26, 2026

Personnel Overview

NAME	FILE NUMBER	DESIGNATION	To Project	From Project	COMPANY	WORK AREA
			Date of Arrival	Date of Dep		
Daryll Enano	5062	CHIEF SURVEYOR	01.07.2025	-	NMDC	Site Office
Mohamad Abror Hediarto	500652	PROJECT SURVEYOR	01.09.2025	-	BRUNEL	Site Office
Danilo Bartolome	502893	PROJECT SURVEYOR	01.12.2025	-	BRUNEL	Site Office
Mateo Mollaneda	503296	SURVEY DATA PROCESSOR	22.12.2025	-	BRUNEL	Site Office
Roderick Clavel	T3274	SURVEYOR	24.03.2025	-	NMDC	Site Office

Leave Planning

NAME	Leave Plans		UAE Visa Run	Leave Type	Remarks	Destination
	Start of Leave	End of Leave				
Daryll Enano	01-12-2026	05-01-2027	09-01-2027	Rotation	Planned	Philippines
Mohamad Abror Hediarto	28.09.2026	24.10.2026	N/A	Annual	Ongoing	Taiwan
Danilo Bartolome	18.12.2026	17.01.2027	N/A	Annual	Planned	Philippines
Mateo Mollaneda	21.01-2027	22.02.2027	N/A	Annual	Planned	Philippines
Roderick Clavel	21-01-2027	21-02.2027	15-01-2027	Rotation	Planned	Philippines

Training and Pass Status

Name	ARC expiry	Offshore Training	OSHA Training date	Taichung Port Gate Pass - Validity	TPC PASS Validity	Remarks
Daryll Enano	2026-11-26	BOSIET 2026-01-20	2026-04-21	2026-06-30	2026-12-31	All Complete
Mohamad Abror Hediarto	2027-08-24	BOSIET 2023-10-20	2025-09-19	2026-06-30	2026-12-31	All Complete
Danilo Bartolome	2027-08-31	Waiting for Project Schedule	2026-01-07	2026-05-30	Application – Processing	Need to wait until Feb 2027 ARC Renewal
Mateo Mollaneda	2026-12-21	BOSIET 2024-01-17	2026-01-16	2026-05-30	Application – Processing	On-going
Roderick Clavel	2027-01-31	BOSIET 2026-04-01	2026-04-21	2026-05-30	2026-12-31	All Complete

(Weekly) General Information

Project Activities with Survey Involvement:

- Survey Base Station at Taichung Port and Tunghsiao Power Plant installed and maintained by EGST. Shared to NMDC via radio telemetry and via NTRIP.

- Tide Gauge installed and maintained by EGST. Shared to NMDC via web page real-time data.

Onshore Taichung, located at Taichung LNG Receiving Terminal:
- Advanced Horizontal Directional Drilling (AHDD) Installation
- Tien-li Pipe Storage area and GRP Coating
- Berth 37 Pipe Storage Area Planning

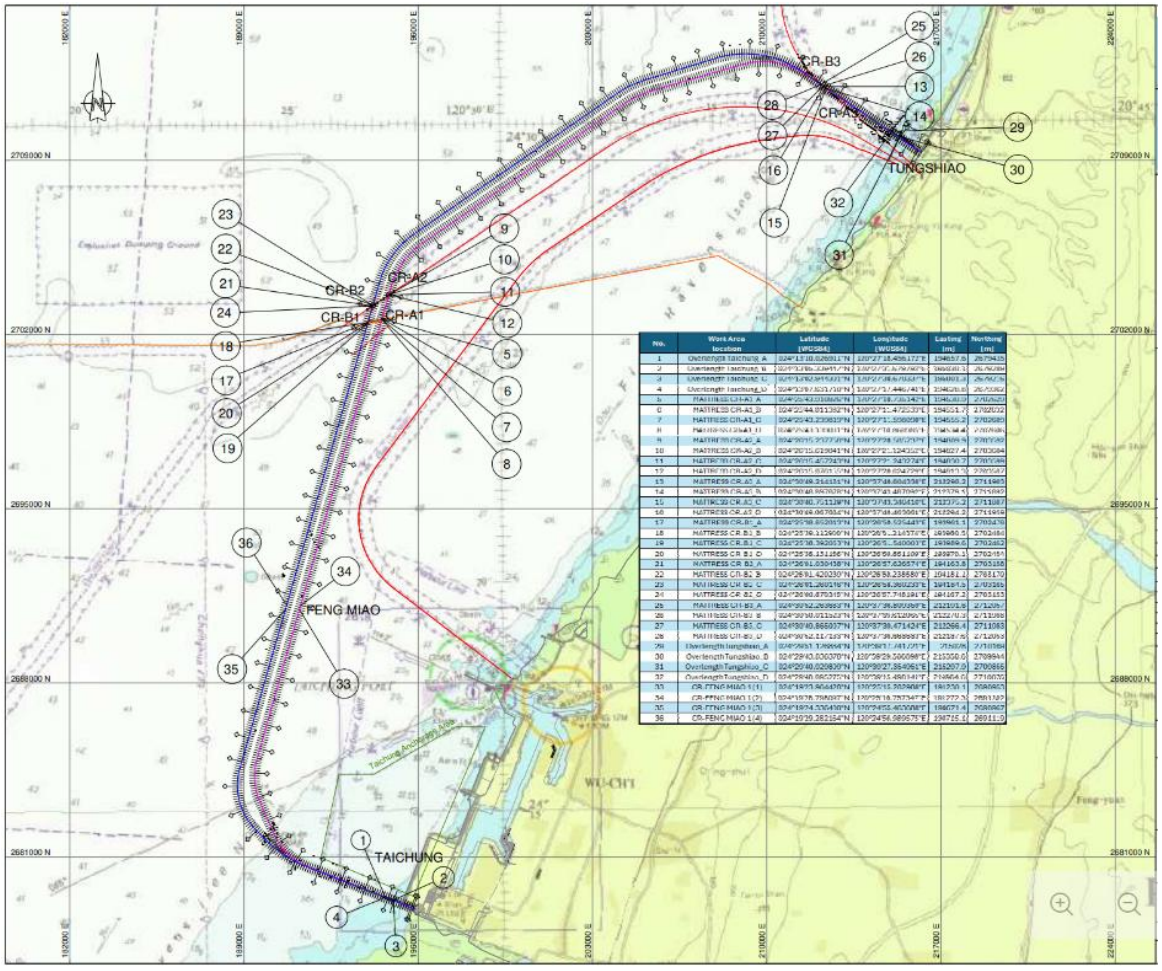
Onshore Tunghsiao, located at Tunghsiao Power Plant
- Temporary Platform Reclamation and Rock Revetment
- Advanced Horizontal Directional Drilling (AHDD) Installation
- Micro tunneling to Power Plant

Nearshore Taichung:
- **Trench 1 + Floatation Channel Dredging works, progress survey, charts and quantity calculations.**

Nearshore Tunghsiao
- **Trench 8 + Floatation Channel Dredging works, progress surveys, charts and quantity calculations.**
- Midline Tie-in Dredging works, progress surveys, charts and quantity calculations.
- LP-1 relocation work

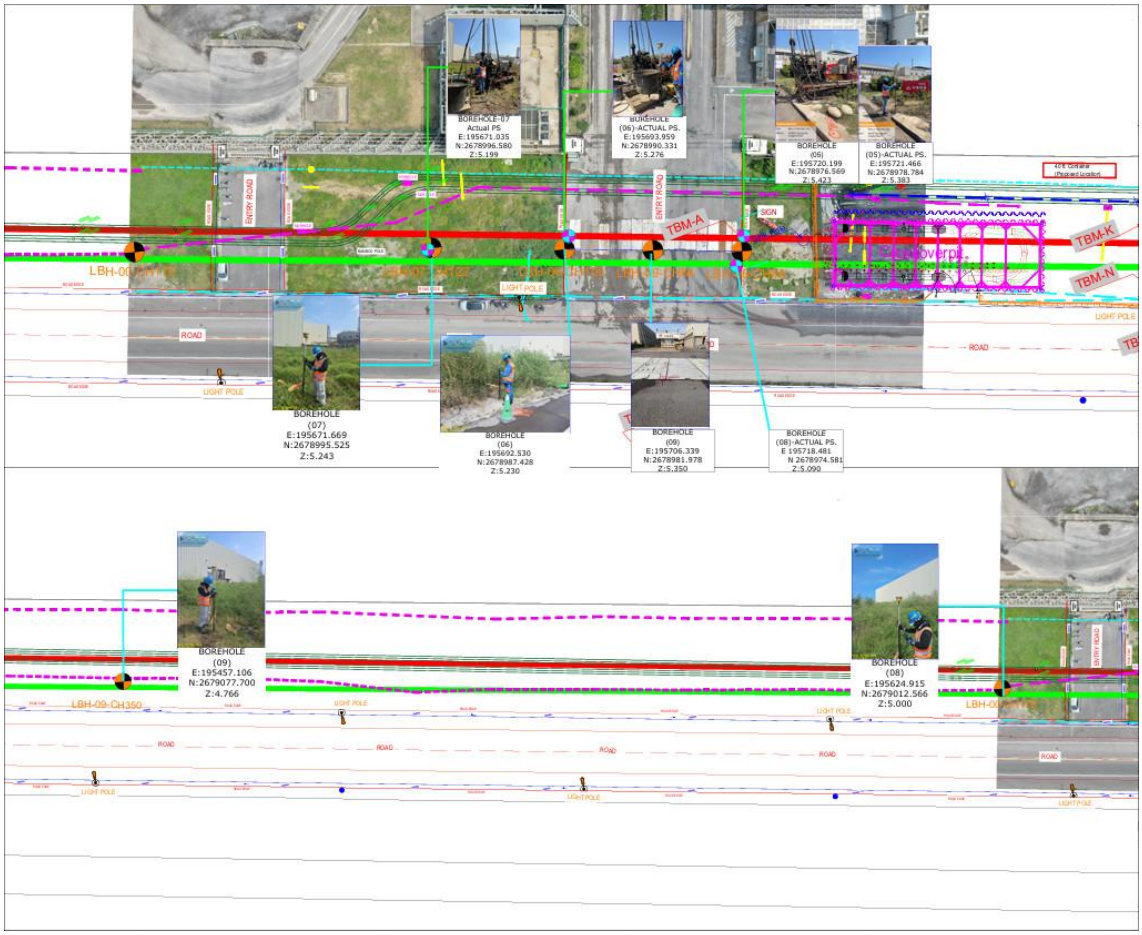
Offshore Work scopes
- **Concrete Mattress Installation on Datan Pipeline Crossings – Pre survey**
- **Concrete Mattress Installation on Yongan Pipeline Crossings – Pre survey**
- **Concrete Mattress Installation on Taijin Telecom Cable Crossings – Pre survey**
- **Concrete Mattress Installation on Fengmiao Windfarm Export Cable Crossings – Pre survey**
- Future Crossings – Yong-an Tonghsiao Pipeline 2 (YT2), Formosa Pipeline, Datan Tonghsiao Pipeline 2 (DT2)
- 54 km (about 33.55 mi) Pipelaying – ICQP design preparations, Pre-lay survey
- Preparation of Plans & Drawings – Progress & Deliverables.

Project Location – Taichung and Tunghsiao - Taiwan

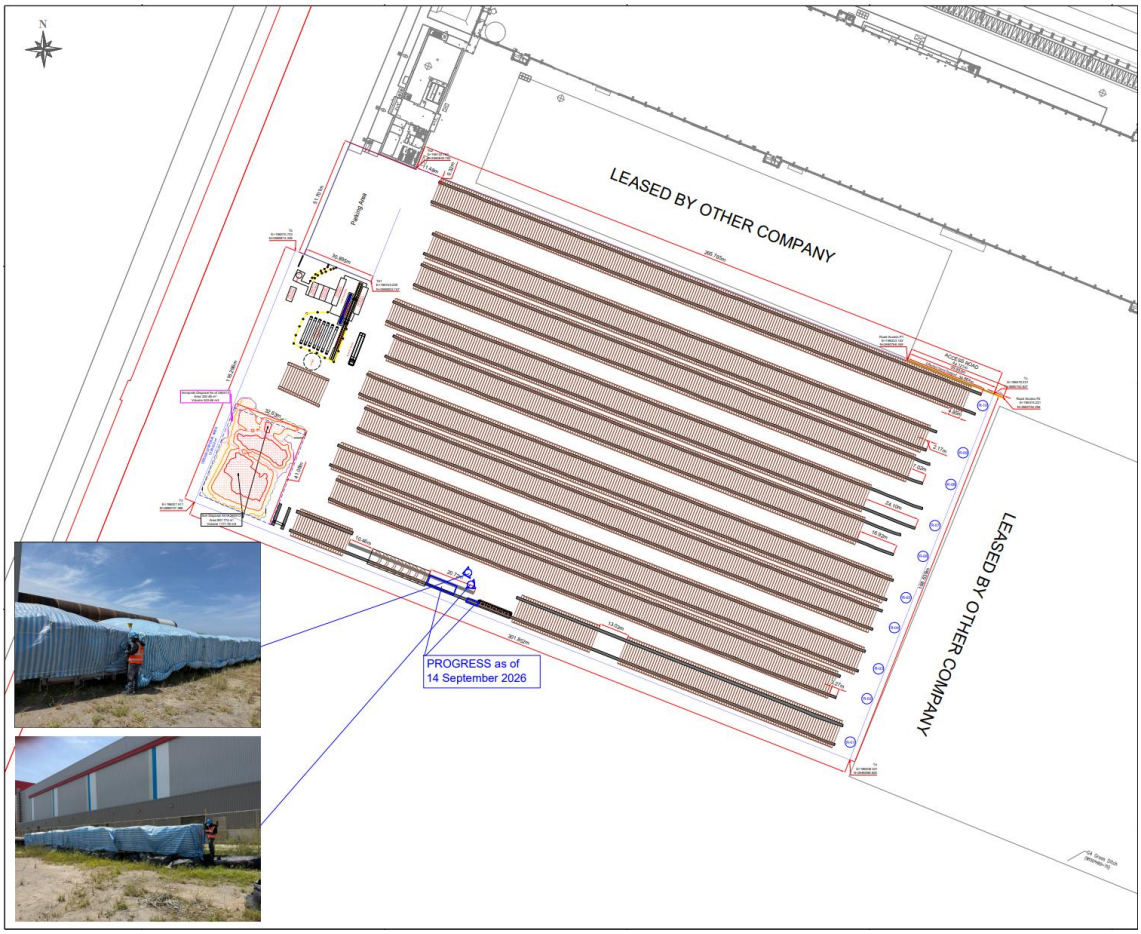


ONSHORE

Taichung AHDD Site: (2026-October-13)

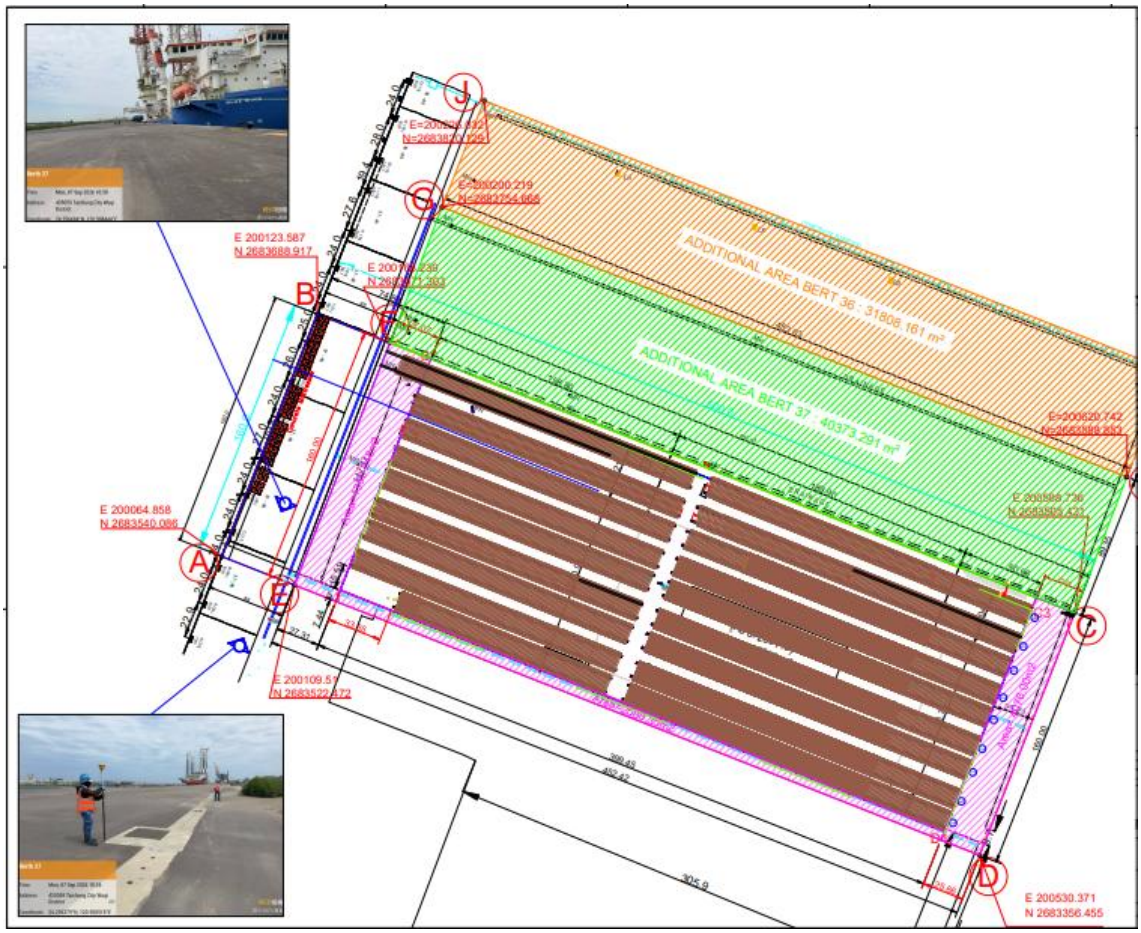


Taichung Tienli Site: (2026-September-20)



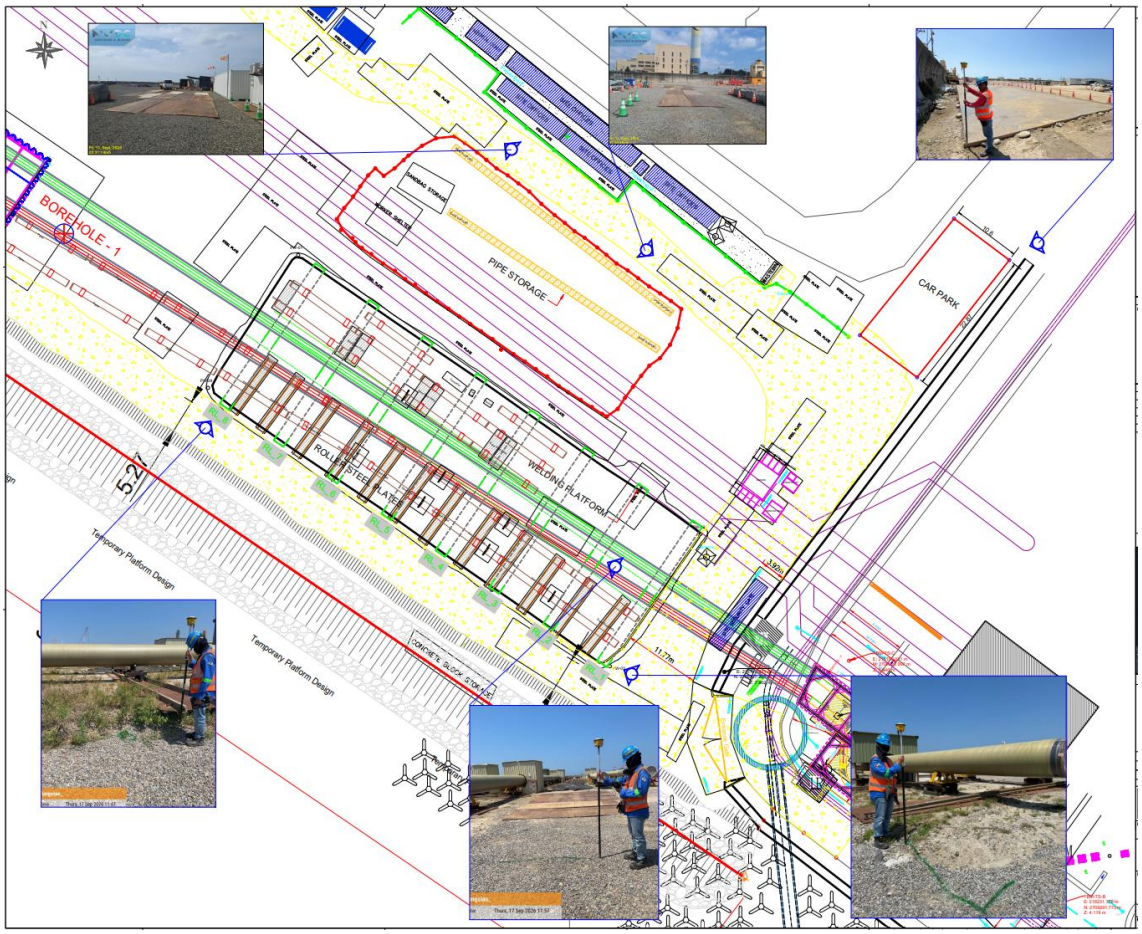


Taichung Berth 37 Site: (2026-September-20)





Tunghsiao Temporary Platform AHDD Site: (2026-September-20)

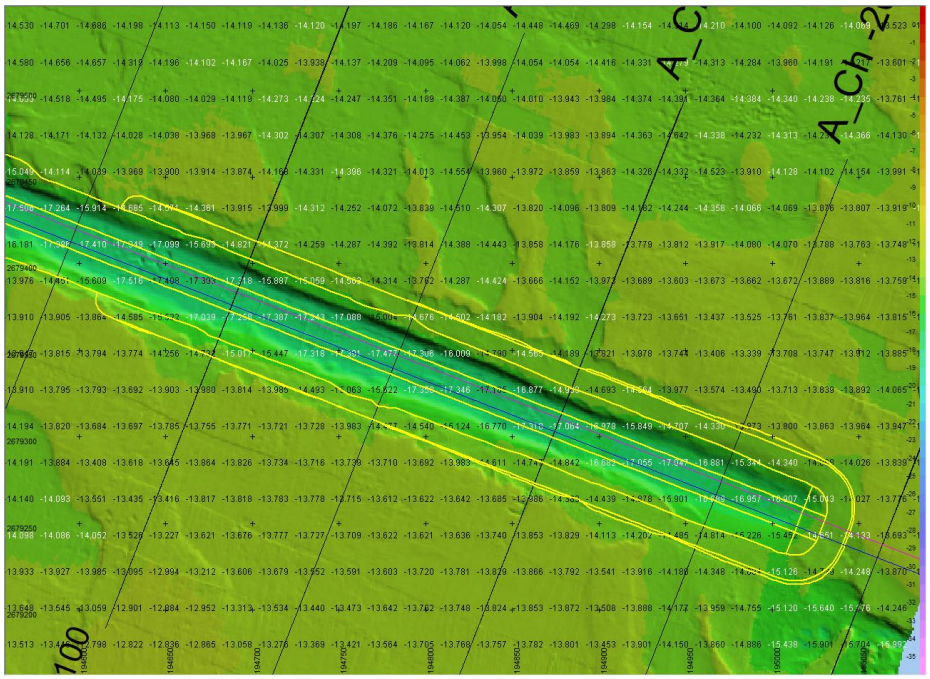




NEARSHORE

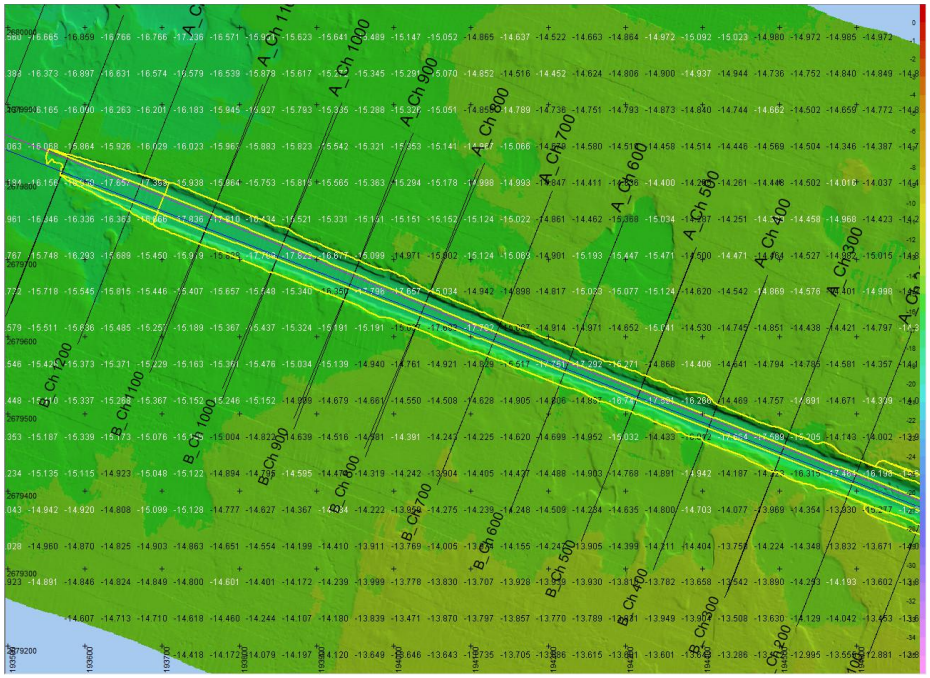
Trench 1 – Floatation Channel CH –265 to +CH 180 (Taichung Nearshore)

– Monitoring Survey (2026-Sept-28)



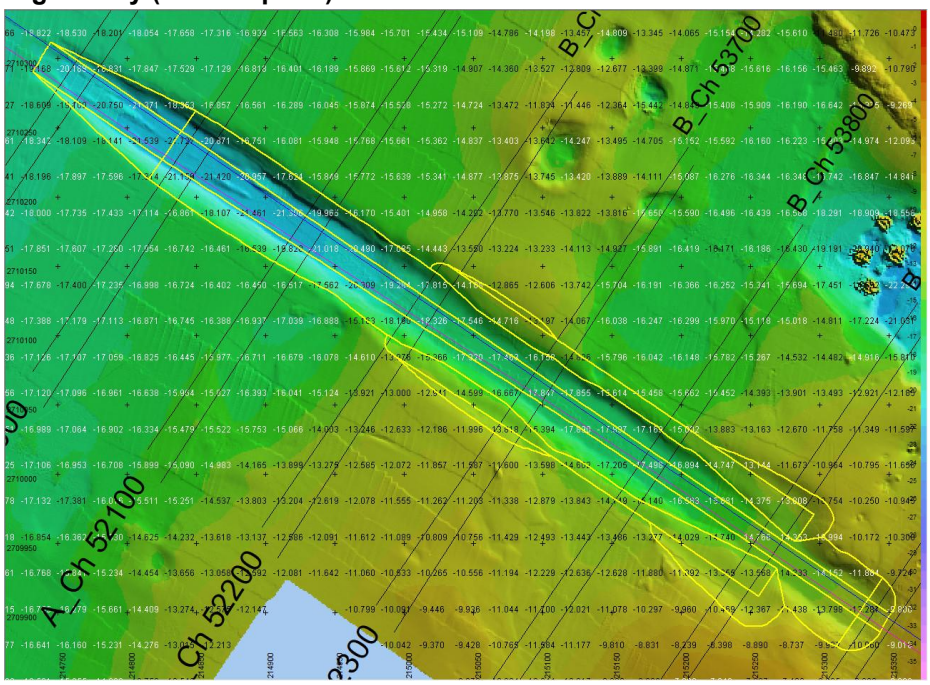
Trench 1 – CH 180 to CH 1340 (Taichung Nearshore)

– Monitoring Survey (2026-Sept-28)



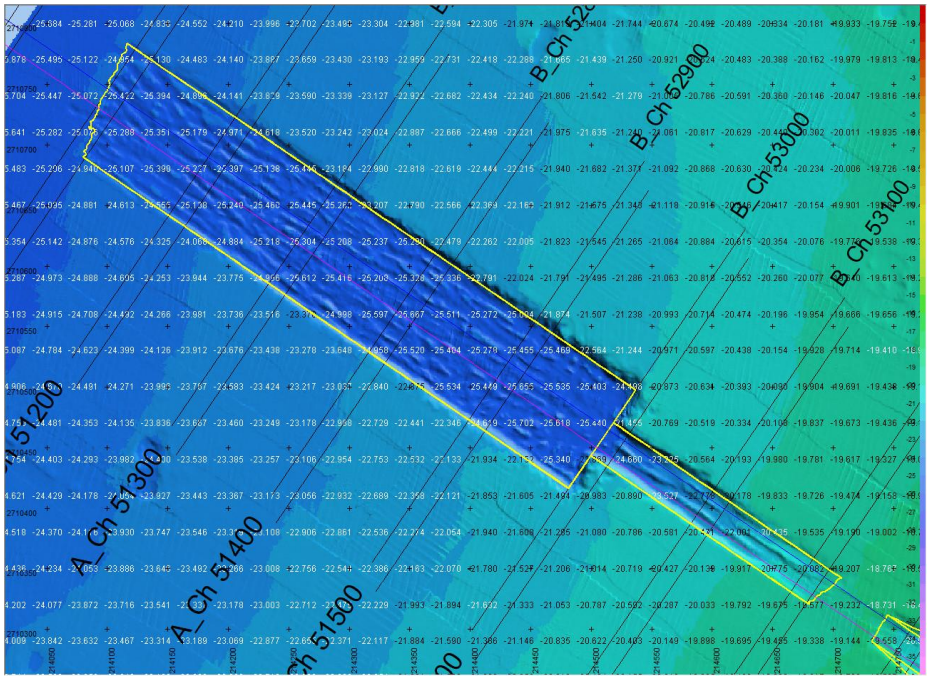
Trench 8 – Ch 53227 to Ch 54010 (Tunghsiao Nearshore)

– Monitoring Survey (2026-Sept-27)



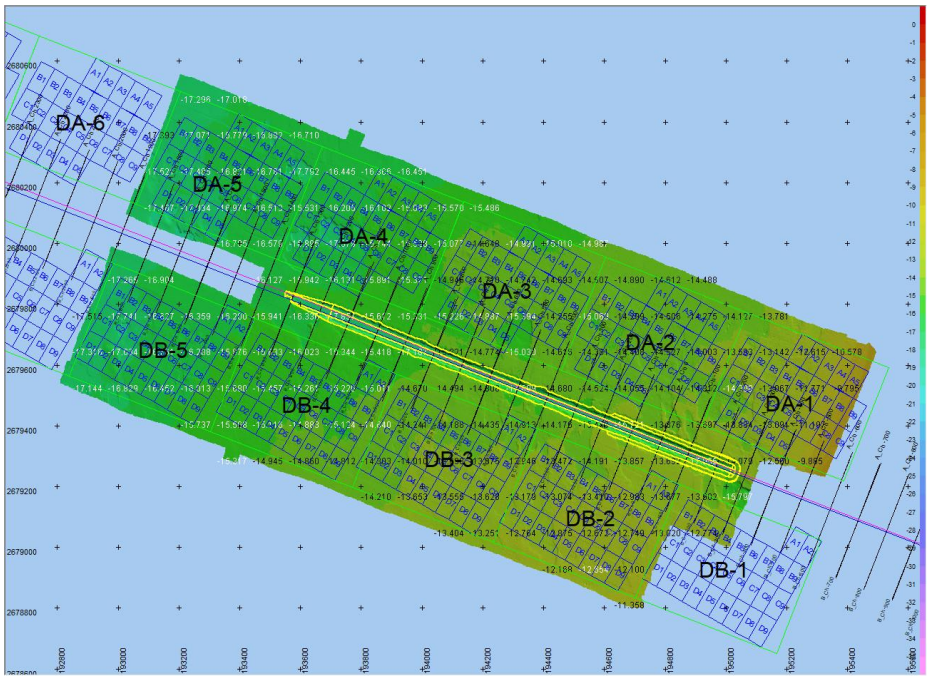
Midline Tie-In – CH 52500 to CH 53221 (Tunghsiao Nearshore)

– Monitoring Survey (2026-September-27)



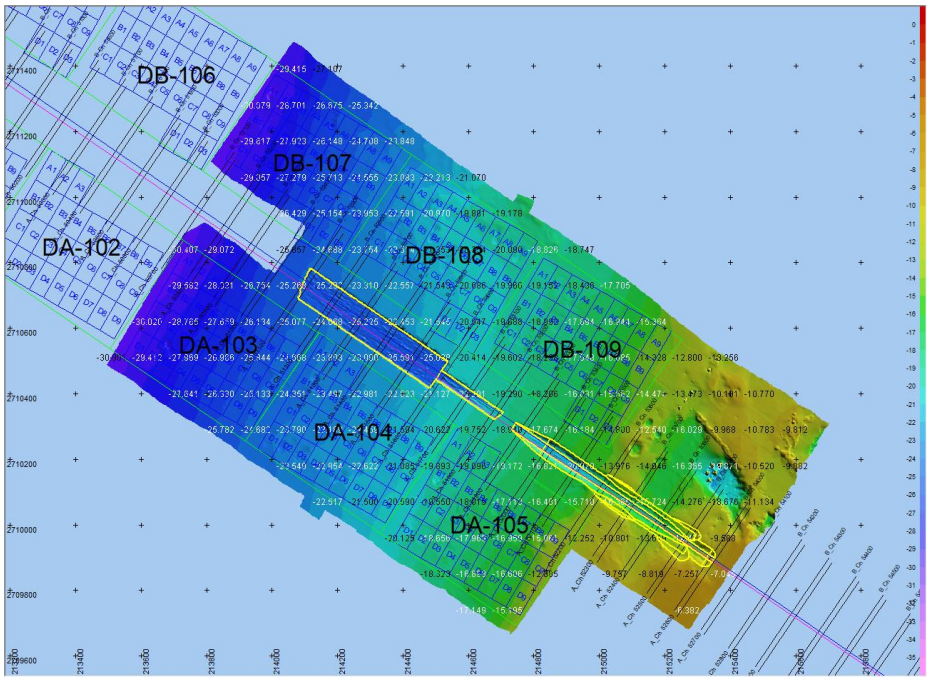
Dumping area monitoring (Taichung Nearshore) (DB 2-5, DA 1-5)

– Monitoring Survey (2026-September-28)



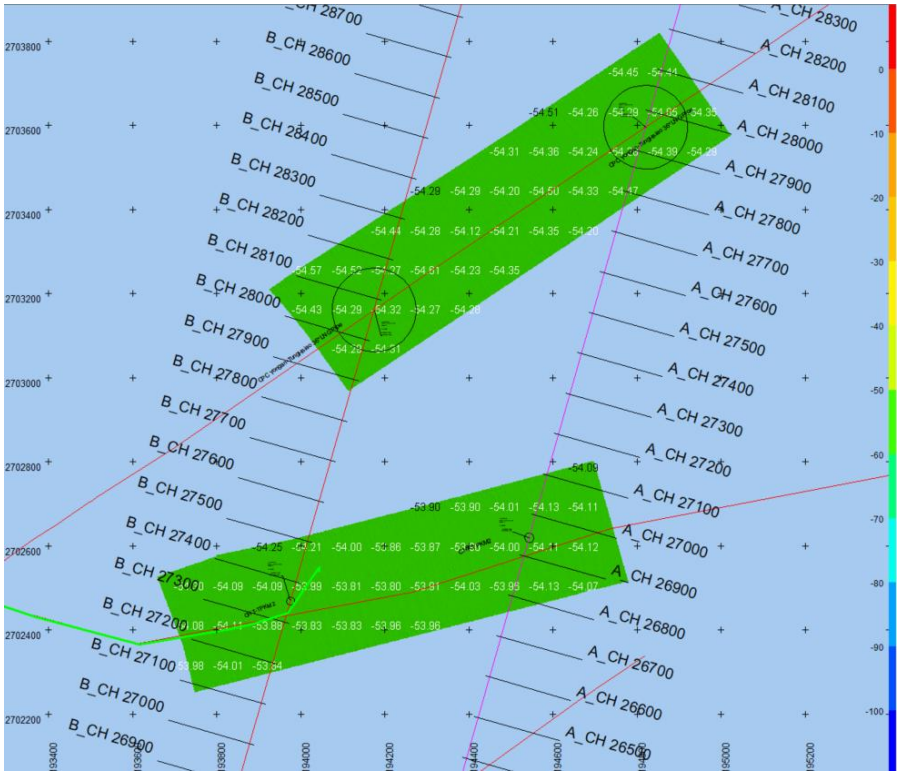
Dumping area monitoring (Tunghsiao Nearshore) (DB 107-109, DA 103-105)

– Monitoring Survey (2026-September-28)

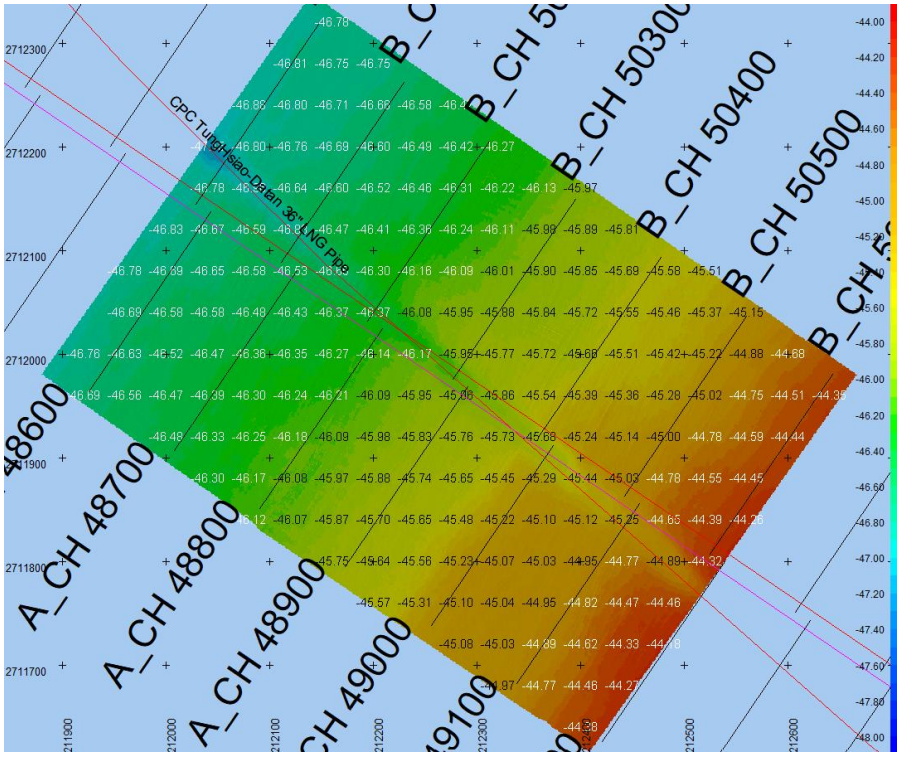


OFFSHORE

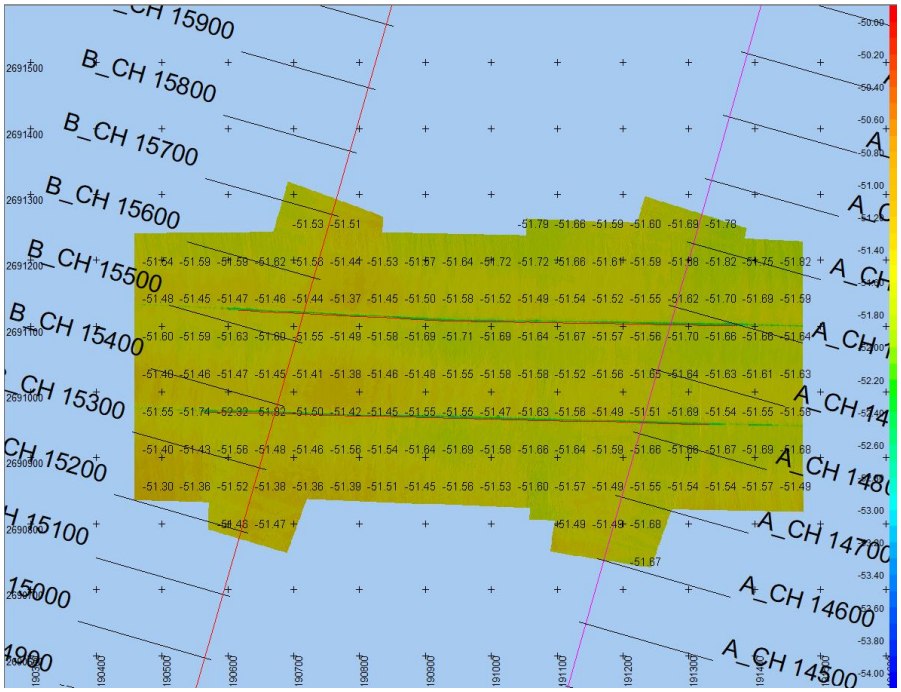
Pre Survey for Concrete Mattress Installation Crossings A 1,2 and B 1,2 (2026-July-29)



Concrete Mattress Installation on Pipeline Crossings A 3 and B 3 (2026-August-17)



Pre-Survey on Pipeline Crossings Fengmiao OEC1 and OEC2 (2026-August-30)



Daily Survey Activities:

19.10.2026 Monday

Onshore-Land: Danilo Bartolome, Mateo Mollaneda, Roderick Clavel
- Taichung AHDD Check Survey - Gather data of AHDD Road settlement monitoring
- Taichung AHDD Survey Reports - Process report of the Taichung AHDD Road settlement report
- Taichung Landside Progress Survey - Generate Sketch Drawing of Taichung Site Weekly Progress Update survey
- Taichung AHDD Stake-Out/Marking - [Thruster Pit] Generate a sketch drawing showing the remarking of the previously marked sheet pile corner near the roadside, with a 0.50 m inward offset from the design location.
- Taichung Landside Survey Input - Drawings - 20261017_RQ0419_Taichung Vertical Drilling Profile

Marine Offshore: Daryll Enano, Mateo Mollaneda
- Offshore Mainline Cross-Section - [DLS-4200] 20261019_RQ0421_Crossing A1-A3 260928 Survey Overview Diff Chart Cross Sections
- Offshore Mainline Cross-Section - [DLS-4200] 20261019_RQ0422_Crossing B1-B2 260928 Survey Overview Diff Chart Cross Sections

20.10.2026 Tuesday

Wuqi Office: Danilo Bartolome
- Taichung Office Internal Survey Reports - Process survey report of the Taichung AHDD Road settlement monitoring

Onshore-Land: Danilo Bartolome, Roderick Clavel
- Tunghsiao Temp Platform Stake-Out/Marking - Assist Thaitan with staking out and marking the location of their facilities area at Tunghsiao Site.
- Taichung Landside Progress Survey - Generate a sketch drawing of the newly installed fence at the Taichung site near the TPC access gate.
- Taichung AHDD Monitoring Survey - Gather data of the Taichung AHDD Road settlement monitoring
- Tunghsiao Temp Platform Benchmark Verification - Conduct traverse levelling for the Tunghsiao platform area TBM for vertical control values and re-evaluation on TBM's interrelationship.

21.10.2026 Wednesday

Wuqi Office: Daryll Enano
- Taichung Office Internal Coordination Meetings - Daily Stand-up Meeting

Onshore-Land: Danilo Bartolome, Daryll Enano, Roderick Clavel
- Taichung AHDD Monitoring Survey - Gather data of the Taichung AHDD Road settlement monitoring
- Taichung AHDD Survey Reports - Generate report of the Taichung AHDD Road settlement monitoring
- Taichung AHDD Progress Survey - [Crossover Pits] Monitor progress of the on-going 2.5m diameter encased Vertical Drilling activity
- Tunghsiao Temp Platform External Meetings - [Microtunneling] Thaitan AHDD Coordination Meeting
- Taichung AHDD Stake-Out/Marking - Stake out and mark the first vertical drilling point of the Thruster Pit, as requested by Nandu. (TH_VD_01)
- Taichung AHDD Stake-Out/Marking - Stake out and mark the alignment of Pipes A and B, as requested by Jeff.
- Tunghsiao Temp Platform Sketch Drawing - Generate a sketch drawing showing the layout and locations of the Thaitan facilities at the Tunghsiao site.
- Taichung AHDD Stake-Out/Marking - Generate a sketch drawing showing the stake-out and marking of the Thruster Pit vertical drilling location (TH_VD-01) at the Taichung site.

Marine Offshore: Daryll Enano, Mateo Mollaneda
- Offshore Mainline Volume Calculation - [DLS-4200] 20261021_RQ0428_Trenching SRI Design Volume 1-5in
- Offshore Subsea Crossings External Meetings - Mattress Installation Meeting

22.10.2026 Thursday

Onshore-Land: Roderick Clavel
- Taichung AHDD Stake-Out/Marking - Generate a sketch drawing showing the stake-out and marking of the Thruster Pit vertical drilling location (TH_VD-13) at the Taichung site.
- Taichung Landside Progress Survey - Update the base drawing using the latest drawings received from the Operations Team, incorporating the latest survey updates for the Tunghsiao site.

23.10.2026 Friday
- No survey tasks logged for this date.

24.10.2026 Saturday
- No survey tasks logged for this date.

25.10.2026 Sunday

Onshore-Land: Danilo Bartolome, Roderick Clavel
- Taichung AHDD Monitoring Survey - Gather data of Taichung AHDD Road settlement monitoring
- Taichung AHDD Survey Report - Process survey report of the Taichung AHDD Road settlement monitoring
- Tunghsiao Temp Platform Survey Input - Drawings - Update the base drawing using the latest drawings received from the Operations Team, incorporating the latest survey updates for the Tunghsiao and Taichung sites.
- Tunghsiao Temp Platform Survey Input - Drawings - Update the Data Controller background using the latest drawings and incorporate the latest survey updates for the Tunghsiao site.

Highlights

1. Taichung Onshore & AHDD Site Operations: Completed daily Taichung AHDD road settlement monitoring surveys and daily processing reports; Generate Sketch Drawing of Taichung Site Weekly Progress Update survey; [Thruster Pit] Generate a sketch drawing showing the remarking of the previously marked sheet pile corner near the roadside, with a 0.50 m inward offset from the design location; 20261017_RQ0419_Taichung Vertical Drilling Profile; Generate a sketch drawing of the newly installed fence at the Taichung site near the TPC access gate; [Crossover Pits] Monitor progress of the on-going 2.5m diameter encased Vertical Drilling activity; Stake out and mark the first vertical drilling point of the Thruster Pit, as requested by Nandu. (TH_VD_01); Stake out and mark the alignment of Pipes A and B, as requested by Jeff; Generate a sketch drawing showing the stake-out and marking of the Thruster Pit vertical drilling location (TH_VD-01) at the Taichung site; Generate a sketch drawing showing the stake-out and marking of the Thruster Pit vertical drilling location (TH_VD-13) at the Taichung site.

2. Tunghsiao Temporary Platform & AHDD Site Operations: Assist Thaitan with staking out and marking the location of their facilities area at Tunghsiao Site; Conduct traverse levelling for the Tunghsiao platform area TBM for vertical control values and re-evaluation on TBM's interrelationship; [Microtunneling] Thaitan AHDD Coordination Meeting; Generate a sketch drawing showing the layout and locations of the Thaitan facilities at the Tunghsiao site; Update the base drawing using the latest drawings received from the Operations Team, incorporating the latest survey updates for the Tunghsiao site; Update the base drawing using the latest drawings received from the Operations Team, incorporating the latest survey updates for the Tunghsiao and Taichung sites; Update the Data Controller background using the latest drawings and incorporate the latest survey updates for the Tunghsiao site.

3. Wuqi Office & Marine Engineering Deliverables: [DLS-4200] 20261019_RQ0421_Crossing A1-A3 260928 Survey Overview Diff Chart Cross Sections; [DLS-4200] 20261019_RQ0422_Crossing B1-B2 260928 Survey Overview Diff Chart Cross Sections; [DLS-4200] 20261021_RQ0428_Trenching SRI Design Volume 1-5in; Mattress Installation Meeting.

Look ahead

1. Request for Diff Charts and Volume calculations for different survey data is expected due to current planning changes and justifications.

2. Update Tunghsiao and Taichung site installation as-built drawings (weekly progress update).

3. Settlement Monitoring (Road and protection sheetpile) at Taichung site for NMDC surveyor (Daily activity).

4. Depth of Burial Survey for Fengmiao Cable Crossings prior to Filter Unit Installation (Under planning with Seatrench and EGST).

5. Scheduled Follow-Up Scope: Stakeout and marking of the vertical drilling point number A_VD_31 at the Taichung site.

6. Scheduled Follow-Up Scope: Generate sketch drawing of the completed stake out and marking of the vertical drilling point location.

Outstanding Tasks

1. TPC Entry Pass for Danilo on hold until 2027 Feb ARC Renewal. Now Danilo can enter site with temporary gate pass, currently Abror, Roderick and Daryll have TPC yearly pass (Taichung and Tunghsiao).

2. Taiwan government E-GNSS correction NTRIP account cannot be requested by NMDC due to company entity registered as foreign company, request to EGST to apply for NMDC operational use. Need to ask EGST for assistance for the EGNSS service. (Stewart Ho assist for the service).

3. Danilo to be scheduled for BOSIET. Waiting for Project Schedule.

4. Rick Jhou - shortlisted as Lead/Chief Surveyor, Sophie update: will join before 23 Nov 2026.

5. Active Field/Office DTR Action Items: Stakeout and marking of the vertical drilling point number A_VD_31 at the Taichung site; Generate sketch drawing of the completed stake out and marking of the vertical drilling point location.

Hardware Movement and Status

1. Survey Forms Compliance - IA-IM-26.005 and Survey Equipment Mob/Demob Compliance - IA-IM-26.006 memo from HQ to be implemented in the Taiwan Project.

NMDC Hardware from Abu Dhabi

No	Eqpt Asset No	Equipment Description	Qty	Notes / Remarks
1	SR140050015S0011	Survey Software - Teledyne-Reson Dongle v90 SRV	1	Personal Daryll
2	SR140030015S0059	Survey Software - Teledyne-Reson Dongle v94 PROC	1	Personal Danilo
3	IT020390315S0017	Lenovo ThinkPad P16S, G1 Laptop	1	AHO from Mok to DRL to be Returned to IT Dept
4	SR140030015S0033	Survey Software - Teledyne-Reson Dongle v94 PROC	1	Personal Mateo
5	SR14003001S0027	Survey Software - Teledyne-Reson Dongle v94 PROC	1	Personal Abror
6	T4NXCV11K133178	Asus Expertbook Laptop	1	Personal Abror
7	N/A	Asus Expertbook Laptop	1	Personal Danilo
8	SR0104000 4S0012	Surface Positioning - Controller Trimble Control Unit - TSC5	1	Rover Set
9	SR010120025S0063	Surface Positioning - GNSS RTK Trimble Receiver - R780 Rover	1	Rover Set
10	N/A	Range Pole	1	Rover Set
11	N/A	Rover Batteries	2	Rover Set
12	N/A	Rover Battery Cradle and adapter	1	Rover Set
13	N/A	Radio Antenna for R780	1	Rover Set
14	N/A	Case for the Rover Set	1	Rover Set
15	SR010400025S0001	Surface Positioning - Controller Trimble Control Unit - TSC3	1	Rover Set
16	SR010120025S0027	Surface Positioning - GNSS RTK Trimble Receiver - SPS985 Rover	1	Rover Set
17	NA	Land Survey - Total Station Leica Reflector - 1203+ with Prism, Cradle, Batteries, Charger, Adapter	1	Land Survey Equipment, 1 TS Battery is broken
18	NA	Rover Batteries, ,	2	Rover Set
19	NA	Rover Battery Cradle and adapter	1	Rover Set
20	NA	Radio Antenna for SPS985	1	Rover Set
21	NA	Case for the Rover Set	1	Rover Set
22	NA	Range pole + Bipod	1+1	Range pole broken
23	NA	Prism Pole	1	Land Survey Equipment
24	NA	Wooden Tripod	2	Land Survey Equipment
25	NA	Aluminum Tripod	1	Land Survey Equipment
26	SR080500025S0015	Leica Sprinter Digital Level	1	Land Survey Equipment
27	NA	PC Workstation	1	Personal Mateo
28	S214804	DTPS Dongle	1	Personal Abror
29	NA	GNSS R780 Lemo Cable	1	Rover Set
30	NA	Leveling Staff Bubble	1	Leveling Set
31	NA	Total station Batteries	3	1 broken, 2 work normal
32	NA	Leica Mini 360 reflector	2	Total Station Set

PDS Dongles

File Number	Name	PDS Dongle Version	PDS Dongle #	Valid till
5062	Daryll Enano	V90	736958_6DE353	02-28-2026
500652	Mohamad Abror Hediarto	V94	736988_6DE321	02-28-2026
502893	Danilo Bartolome	V94	736971_6DE350	02-28-2026
503296	Mateo Mollaneda	V94	738754_6DED74	02-28-2026

NMDC Survey Vessel & Chartered Survey Vessel

VESSEL NAME	TYPE	COMPANY	LOCATION
Woen Sheng No.166 (WS166)	Survey Boat	EGST	Taiwan Strait, Taichung
ECHO 4	Survey Boat	SeaTrench	Taiwan Strait, Taichung

Dredging and Reclamation Overall Quantity

Survey Date	Scope	Design (m3)	Progress (m3)		Remark
			Within Design	Overall	
20 Apr 2026	Nearshore Dredging	476,515.04	454,539.60	538,517.27	Post Survey
13 Jan 2026	Tunghsiao Reclamation	69,641.6	69,641.60	76,934.66	Post Survey

Survey Document Summary

Date Issue	Rev.	Status	DOC No.	Client No.	Title	Document Type
3/3/2025	A	Returned for correction	2913-PP-DM-101	TPSP-TK01-PLN-MDC-0005	Submarine Gas Pipeline Route Delineation (Survey) Plan, Method Statement For Geophysical, Underground Detection Survey, Bathymetric And Topographic Survey	Plan / MS
17/3/2025	B	Returned for correction				
28/3/2025	C	Returned for correction				
17/4/2025	0	Returned for correction				
1/6/2026	0A	Returned for correction				
13/6/2026	0B	No comment				
24/9/2026	1	Others (Approved)				
7/7/2025	A	Returned for correction	2913-PP-DM-108	TPSP-TK01-REP-MDC-0003	GEOPHYSICAL Investigation, Subsurface Detection Survey, Bathymetric and Topographic Survey Report	Report
13/8/2025	B	Returned for correction				
13/10/2025	C	No comment				
4/11/2025	0	Others (Approved)				
28/7/2025	A	Returned for correction	2913-PP-DM-109	TPSP-TK01-REP-MDC-0010	Submarine Gas Pipeline Route Delineation (Survey) Report – Onshore, Nearshore, and Offshore Geological Survey Report	Report
16/9/2025	B	No comment				
12/11/2025	0	Approved, proceed with fabrication				
24/7/2025	A	Returned for correction	2913-PP-DM-114	TPSP-TK01-REP-MDC-0009	Underwater Cultural Heritage Survey Plan – Route Investigation: Supplemental Survey	Plan / MS
12/8/2025	B	No comment				
12/9/2025	0	Others (Approved)				
8/10/2025	A	Returned for correction	2913-PP-DM-115	TPSP-TK01-REP-MDC-0011	Underwater Cultural Heritage Survey Report – Route Investigation-	Report
	B	Revison ongoing				
8/12/2025	A	Returned for correction	2913-PP-OF-205	TPSP-TK01-SD-REP-0014	Pre-Dredge Survey Report	Report
	B	Revison ongoing				
24/12/2025	A	Returned for correction	2913-PP-OF-010	TPSP-TK01-SD-REP-0044	Pipelay Positioning Survey Procedure	Plan / MS
5/2/2026	B	Returned for correction				
17/3/2026	C	No comment				
8/4/2026	0	Others (Approved)				
26/1/2026	A	Returned for correction	2913-PP-DM-116	TPSP-TK01-REP-MDC-0012	Survey Report for Pre-Construction Bathymetry Survey for Dredging and Backfilling	Report
23/2/2026	B	Returned for correction				
9/1/2026	C	Minor comment				
15/09/2026	0	Issued for approval				
24/9/2026	1	Others (Approved)				
31/3/2026	A	Returned for correction	2913-PR-DM-001	TPSP-TK01-REP-SR 0001	Verification of Control Points Survey Report - March-2026	Report
9/4/2026	B	No comment				
24/09/2026	0	Issued for approval				
6/5/2026	A	Returned for correction	2913-PP-DM-118	TPSP-TK01-REP-MDC-0014	Post-Dredge Bathymetry Survey Report for Dredger ARZANA Operation of Dredging	Report
25/5/2026	B	Returned for correction				
18/6/2026	C	No comment				
8/7/2026	0	Others (Approved)				
2/7/2026	A	Returned for correction	2913-PR-DM-002	TPSP-TK01-REP-SR 0002	Verification of Control Points Survey Report - June-2026	Report
31/7/2026	B	No comment				
20/8/2026	0	Others (Approved)				
14/8/2026	A	Returned for correction	2913-PP-DM-119	TPSP-TK01-REP-MDC-0015	TWTK2 Cable Crossing and CPC Pipeline Crossing Burial Inspection Report	Report
16/09/2026	B	Issued for review				
30/9/2026	0	Issued for approval				
6/10/2026	A	No comment	2913-PR-DM-003	TPSP-TK01-REP-SR 0003	Verification of Control Points Survey Report - Sept-2026	Report
6/10/2026	0	Issued for approval				

Base Station and Tide Station:

- EGST Taichung RTK BASE STATION FREQUENCY = 455.0000MHz @ Taichung Port.
- EGST Tunghsiao RTK BASE STATION FREQUENCY = 459.5000MHz @ Tunghsiao Power Plant.
- EGST TIDE WEB Server: Link: 211.20.255.60:6500
  Username/Password: egst01 / egst01
- EGST TIDE Base Station Ntrip: 211.20.255.60: 6000
  Username/Password: egst01 / egst01, egst02 / egst02, egst03 / egst03, egst04 / egst04
  Mount Point: EGST_TCP, EGST_TXO

Name: Daryll Enano

Chief Surveyor

Project: Tung-Hsiao Power Plant 2nd Stage Renewal Project Turn-key Contract of Subsea Gas Pipeline

Cost Code: 20010

Date: 26th October 2026